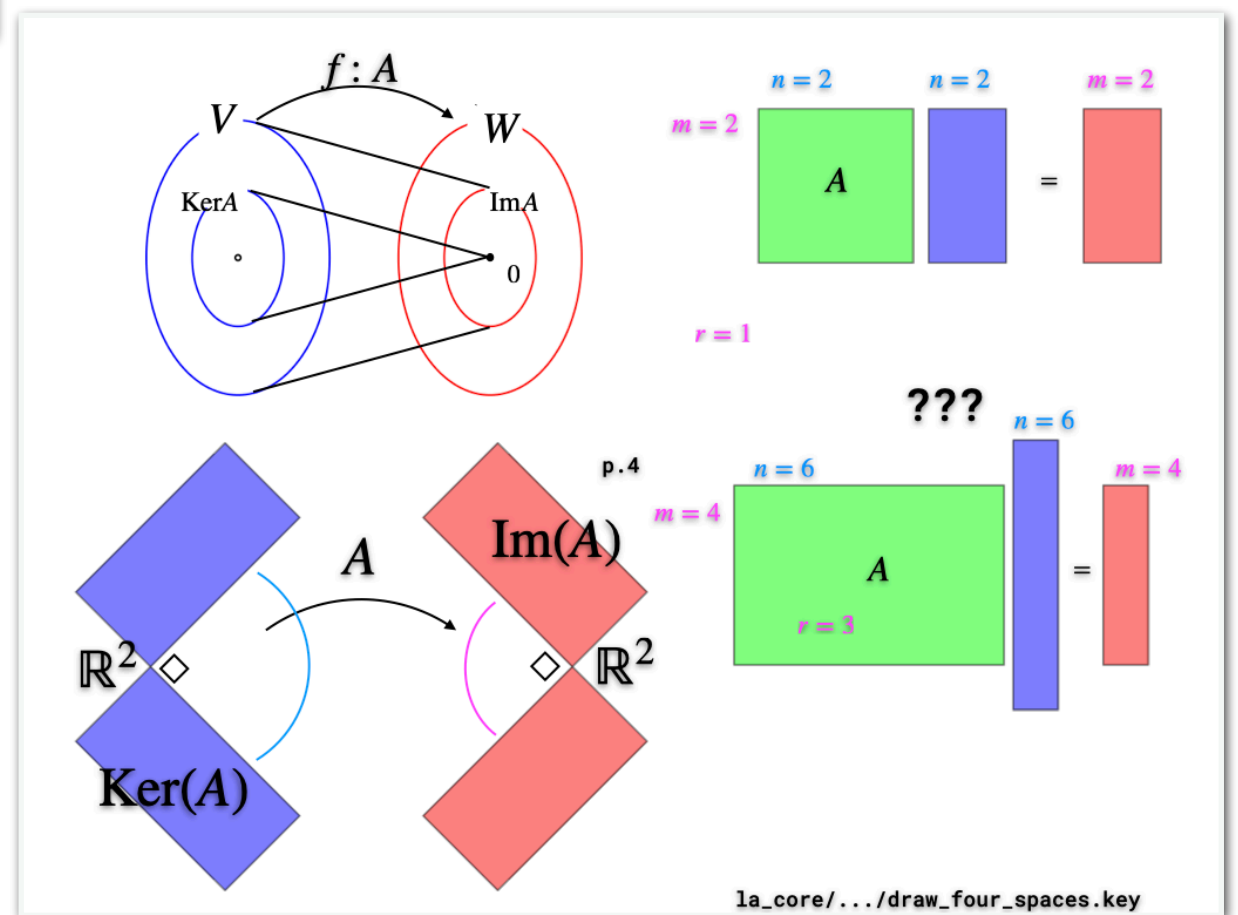
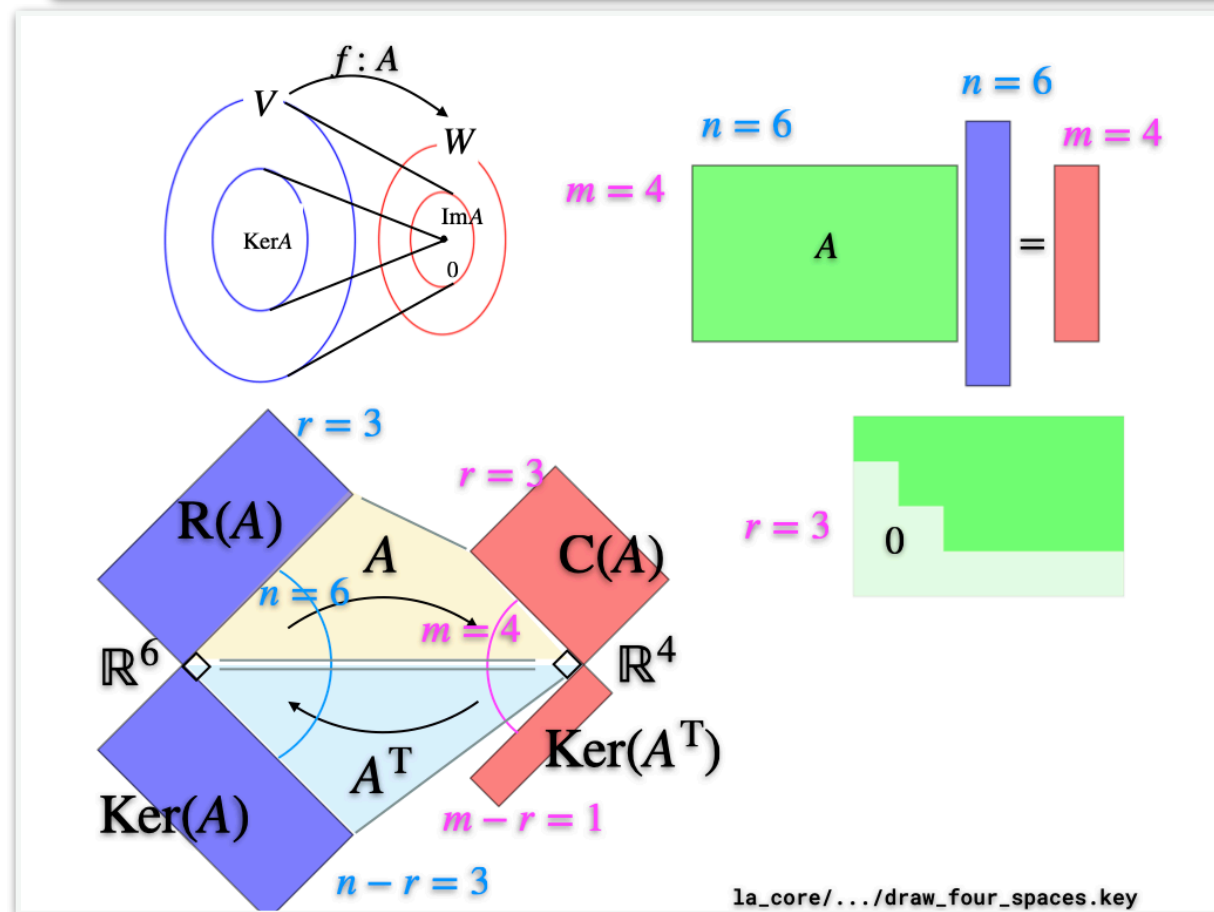
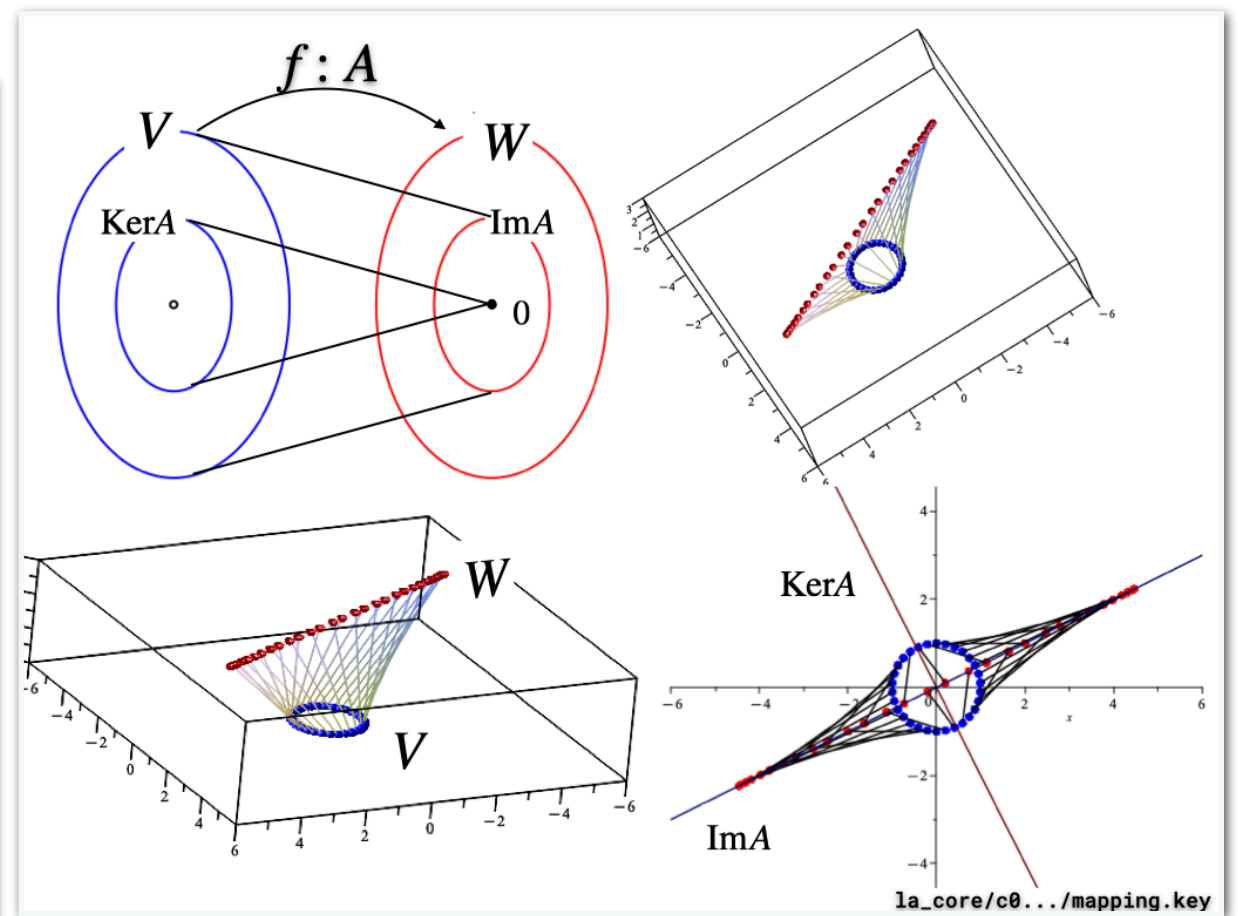
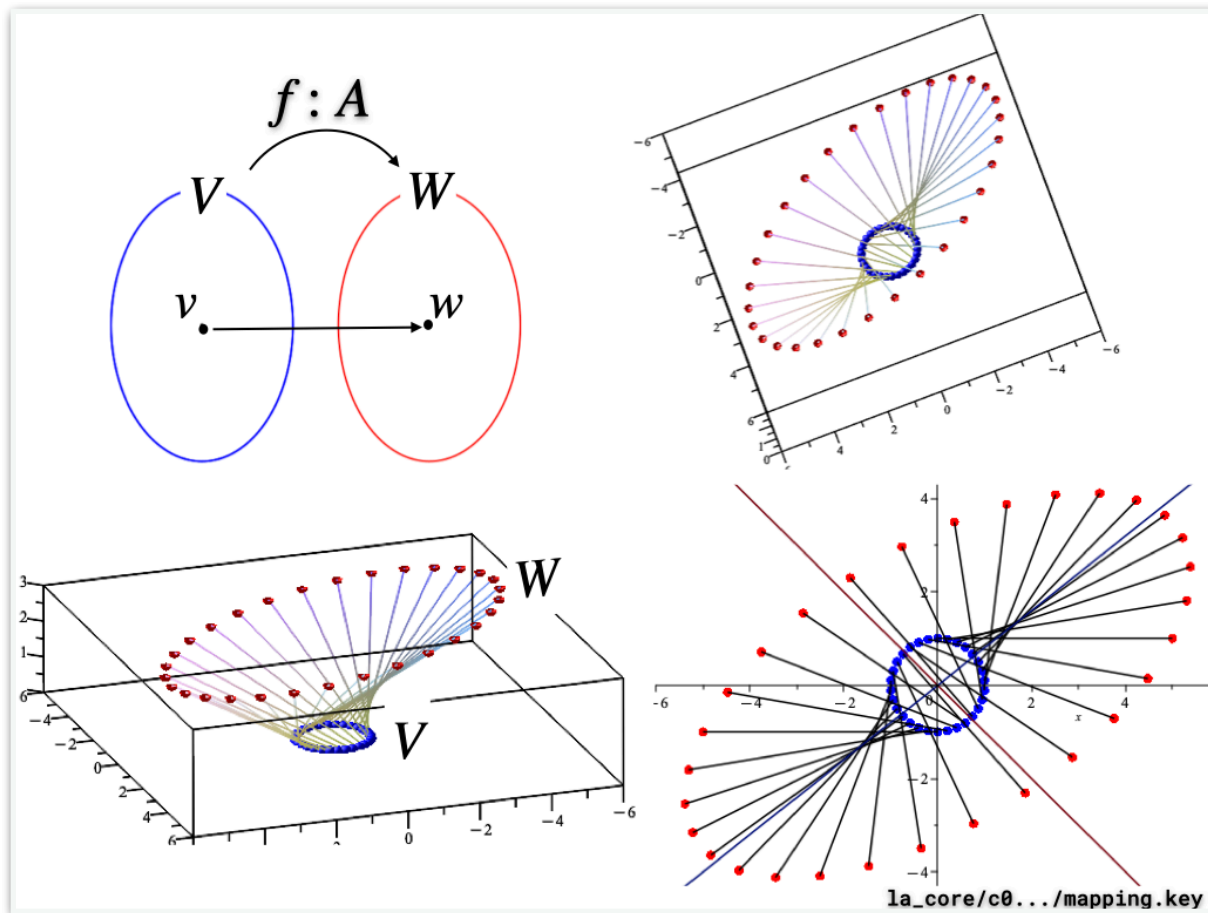


線形代数のイメージ

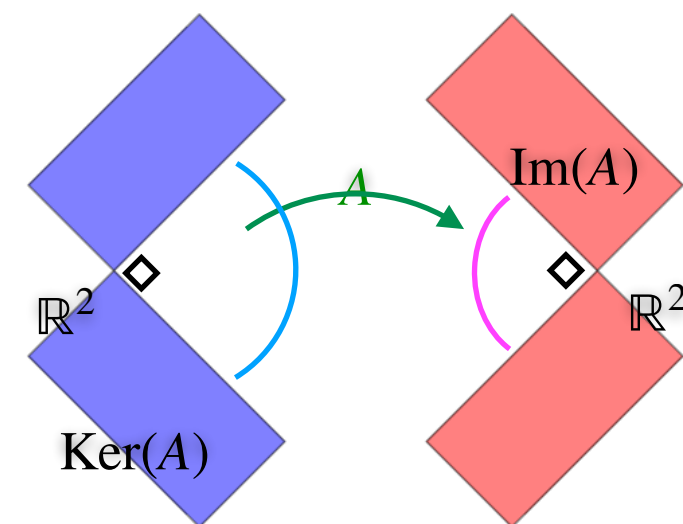
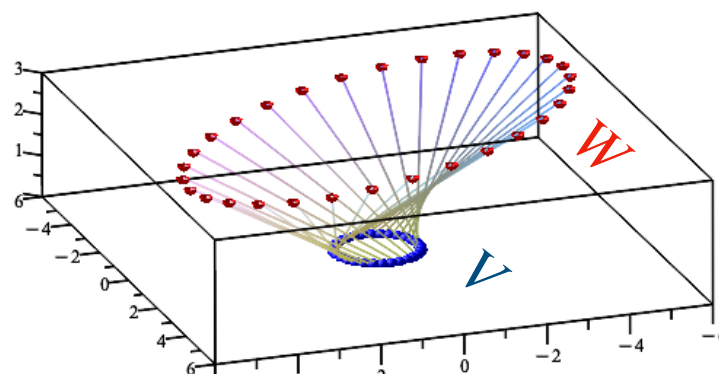
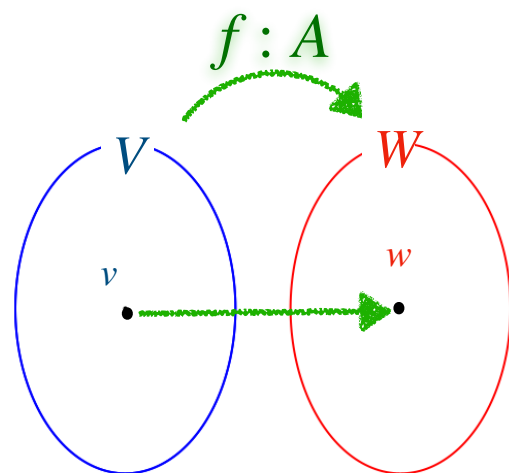
keyword: column space, row space, null space, QR decomposition

source: la_core_26s(GoodNote), la_cr_basic.ipynb



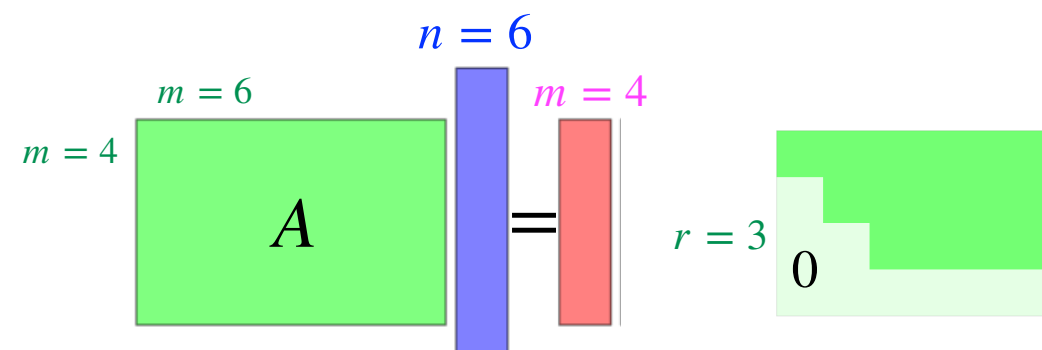
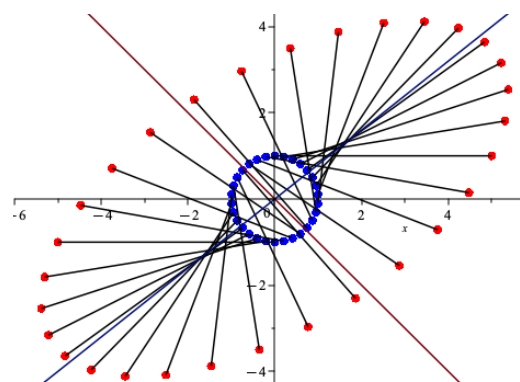
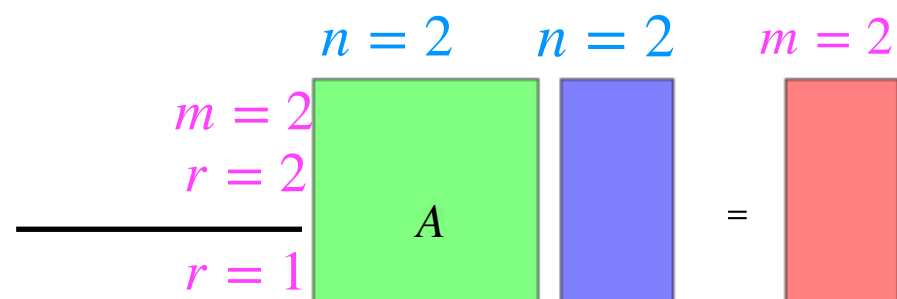
写像(Mapping)

$$\begin{pmatrix} 2 & 5 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} w_x \\ w_y \end{pmatrix}$$



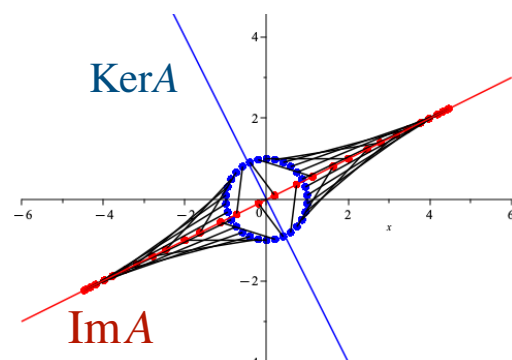
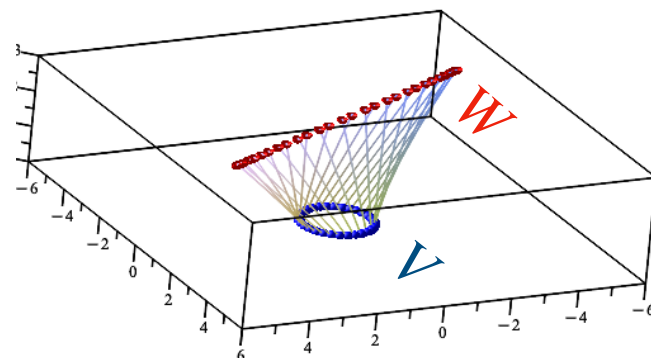
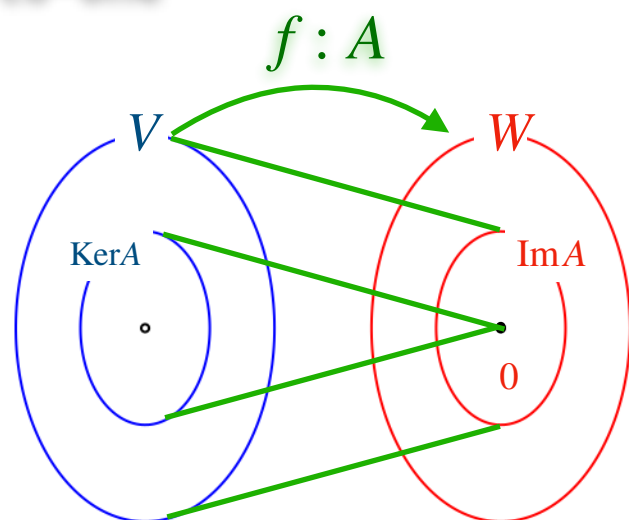
onto 全射

one-to-one 单射

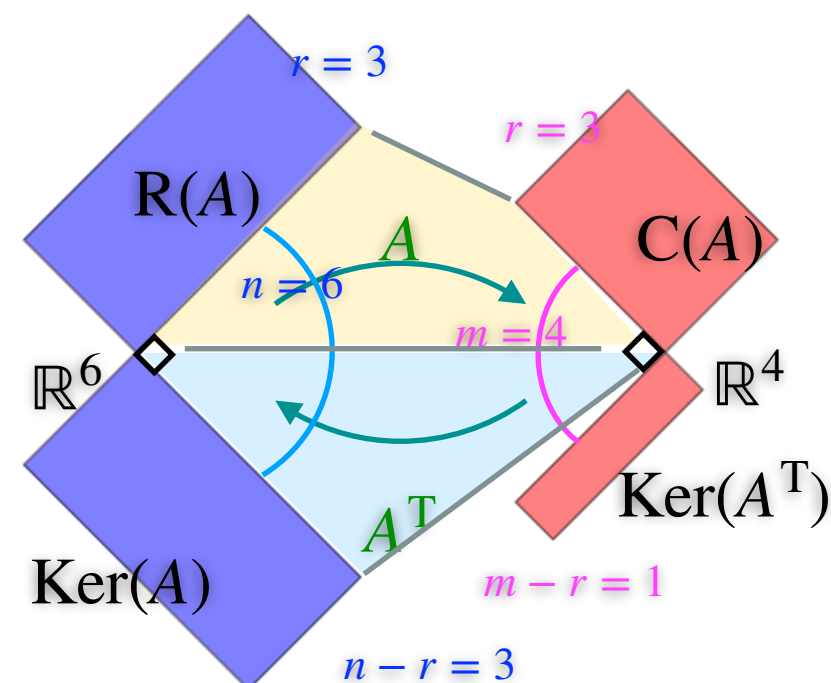


not onto

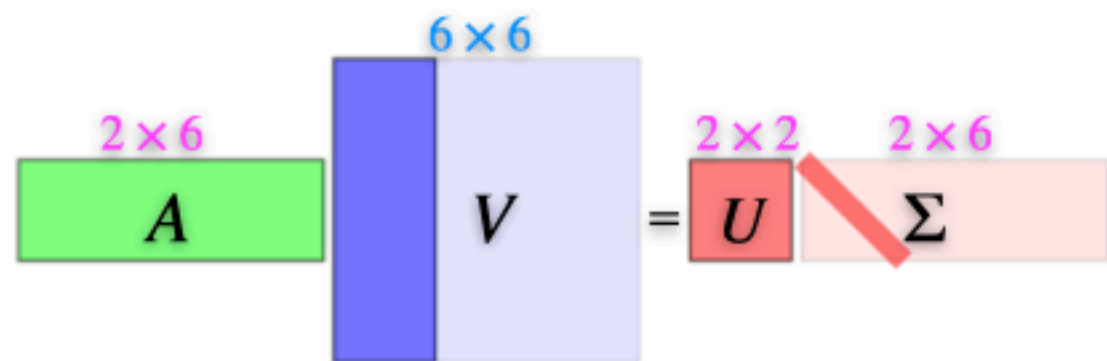
not one-to-one



$$\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} w_x \\ w_y \end{pmatrix}$$



主成分分析(特異値分解)



```
> a1:=Column(U,1)*dS[1][1].Row(Vt,1);
```

$$a1 := \begin{bmatrix} 4.1926 & -4.0426 & 5.9140 & -0.1499 & -1.7215 & -4.1926 \\ 6.1926 & -5.9711 & 8.7352 & -0.2215 & -2.5426 & -6.1926 \end{bmatrix}$$

```
> a2:=Column(U,2)*dS[2][2].Row(Vt,2);
```

$$a2 := \begin{bmatrix} -1.1926 & 0.0426 & 1.0860 & 1.1499 & -2.2785 & 1.1926 \\ 0.8074 & -0.0289 & -0.7352 & -0.7785 & 1.5426 & -0.8074 \end{bmatrix}$$

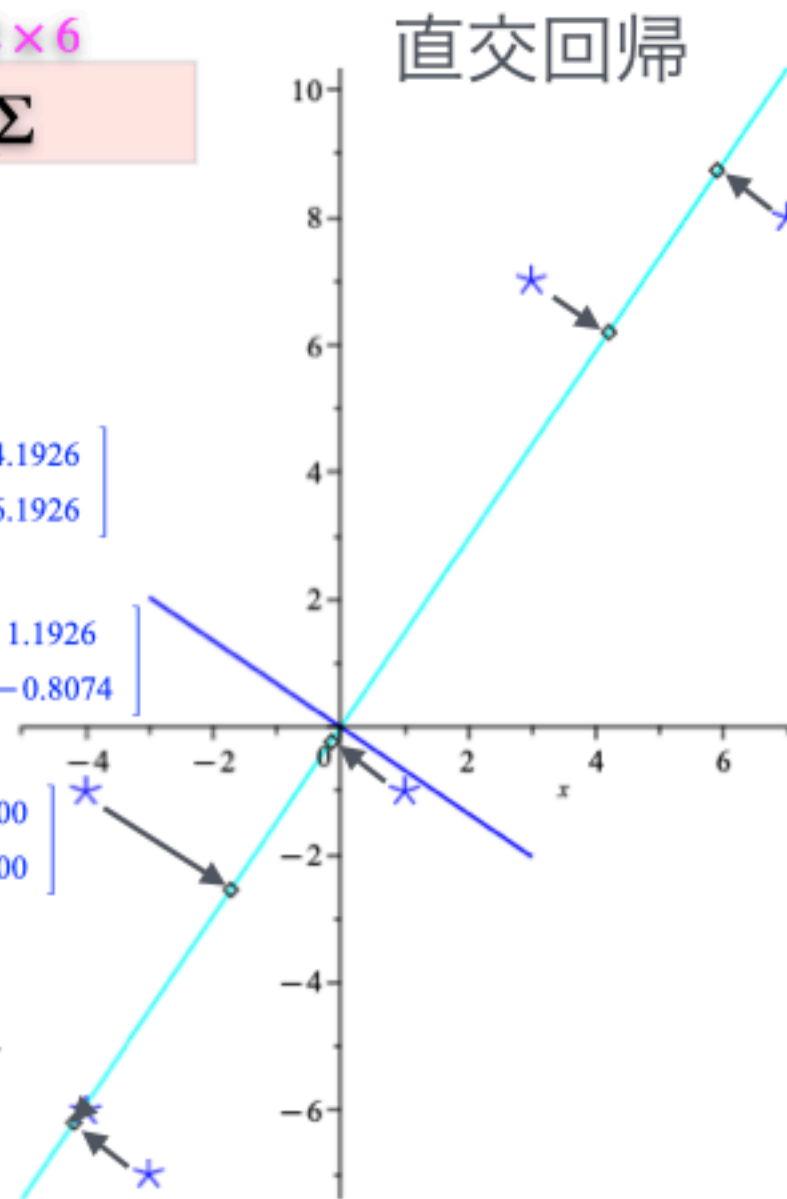
```
> a1+a2;
```

$$\begin{bmatrix} 3.0000 & -4.0000 & 7.0000 & 1.0000 & -4.0000 & -3.0000 \\ 7.0000 & -6.0000 & 8.0000 & -1.0000 & -1.0000 & -7.0000 \end{bmatrix}$$

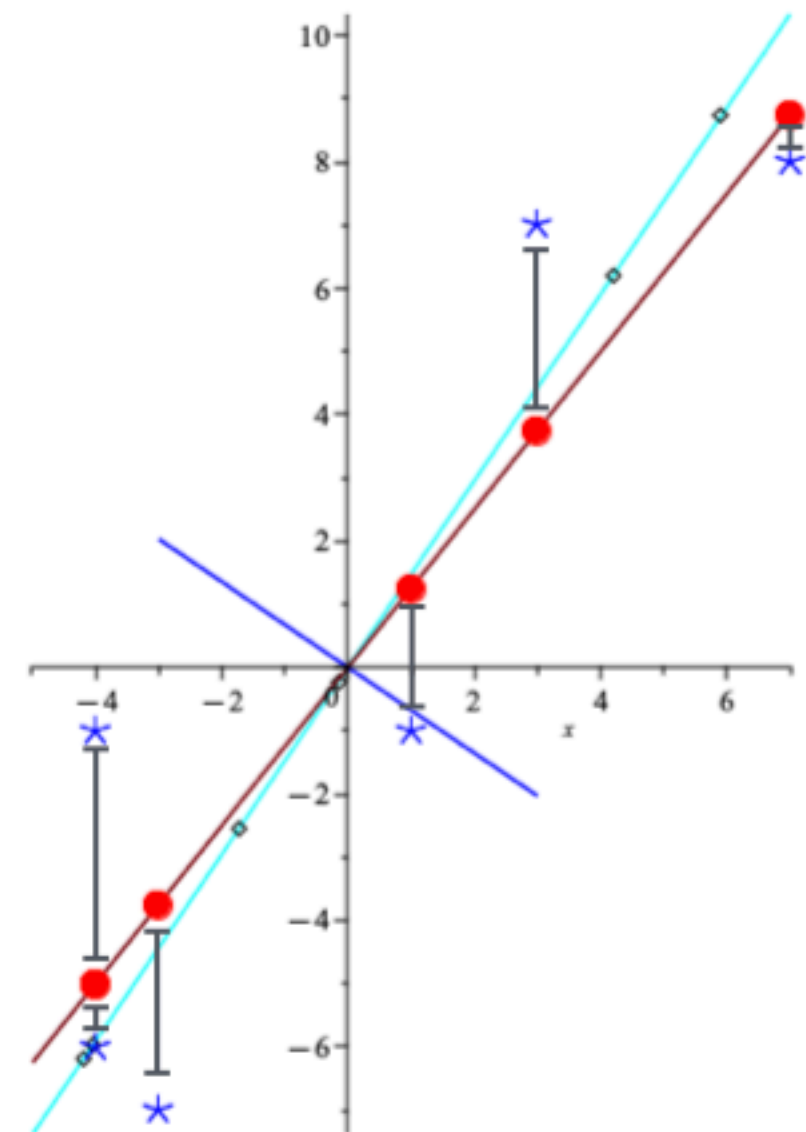
データ点からu1方向の直線
への距離の2乗和が最小

$$\|A - A_k\| < \|A - B\|$$

Eckart-Young



最小二乗法(QR分解)



$$\min \|Ax - b\|^2$$

or

$$\|A\hat{x} - b\|^2 \leq \|Ax - b\|^2$$